

Unit-4
Lecture-3

Quantum tunnelling effect
On Alpha particle decay

Rectangular Potential Barrier $\Rightarrow$ (Quantum Theory)

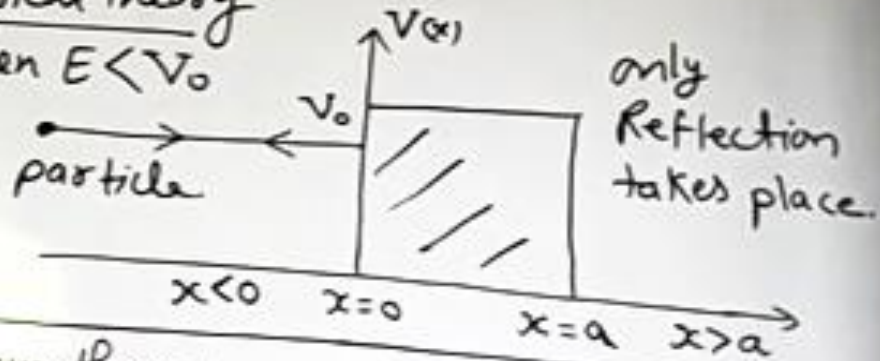
Tunneling $\Rightarrow$

or Tunnel Effect $\Rightarrow$

If a particle with energy E is incident on a potential barrier of height greater than E , then there is a finite probability of the particle, to penetrate the barrier and continue its forward motion. This probability of crossing the barrier is called the Tunnel Effect, because the particle crosses the barrier without going over top but it has passed through a tunnel in the Potential Barrier.

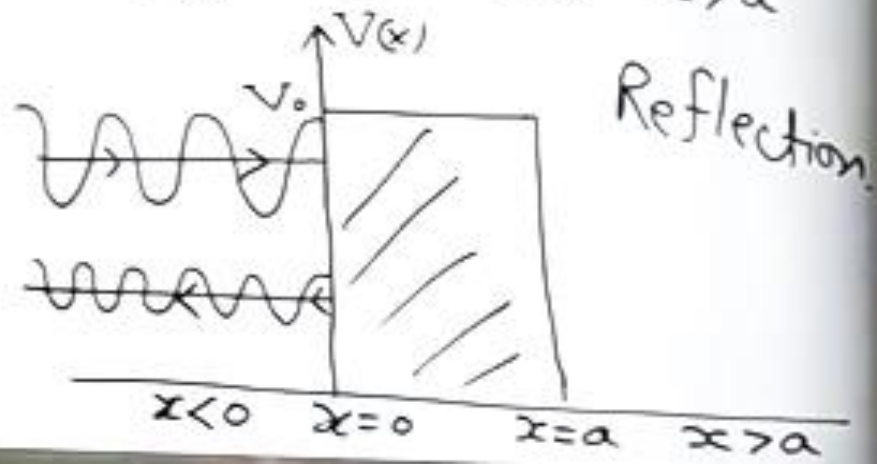
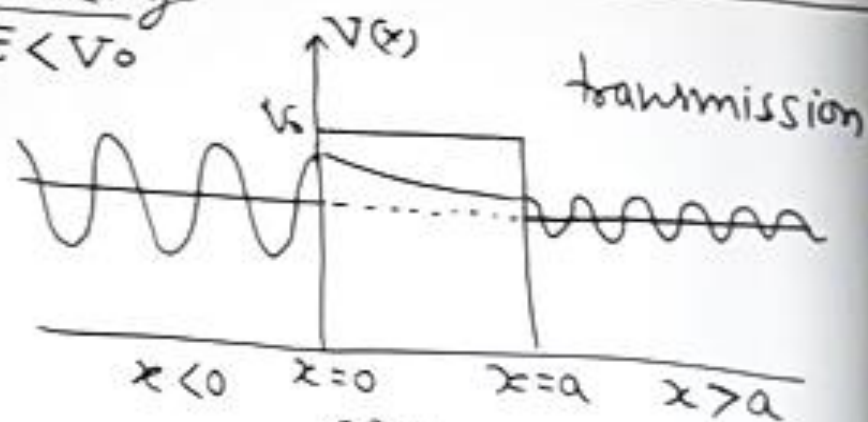
Classical theory

When $E < V_0$



Quantum theory

When $E < V_0$



Quantum tunneling (Tunnel effect) $T \approx e^{-2\kappa L}$ $\kappa = \frac{\sqrt{2m(V-E)}}{\hbar}$ $L = \text{Barrier Width}$

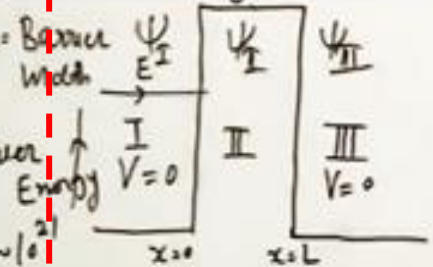
1) There is a finite probability of penetrating the barrier even particle have less Energy than barrier height.

2) Classically phenomena is not possible. $v = \frac{v}{2R_0} \sim \frac{1.07}{10^{-10}} \sim 10^7$

Quantum mechanical - A Ray of hope even $E < V$ little finite probability of penetration $\lambda = \frac{v}{T}$

3) In Case of α particle decay 2.5 MeV height barrier is penetrated by α particle having Energy between 4 to 9 MeV α particle have strike the Wall 10^{-14} m as mass

from Schrodinger Equation $\frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2}(E-V)\psi = 0$ (α decay)



Boundary Condition

$\psi_I = \psi_{II}$
 $\frac{\partial \psi_I}{\partial x} = \frac{\partial \psi_{II}}{\partial x}$ at $x=0$

$\psi_{II} = \psi_{III}$
 $\frac{\partial \psi_{II}}{\partial x} = \frac{\partial \psi_{III}}{\partial x}$ at $x=L$

$A+B = C+D$

Now put $\kappa_1 = \frac{\sqrt{2mE}}{\hbar}$ and $\kappa_2 = \frac{\sqrt{2m(V-E)}}{\hbar}$

$= \frac{\hbar}{\hbar} = \frac{2\pi}{\lambda}$

then $\frac{\partial^2 \psi_I}{\partial x^2} + \kappa_1^2 \psi_I = 0$
 and $\frac{\partial^2 \psi_{III}}{\partial x^2} + \kappa_1^2 \psi_{III} = 0$
 and $\frac{\partial^2 \psi_{II}}{\partial x^2} - \kappa_2^2 \psi_{II} = 0$

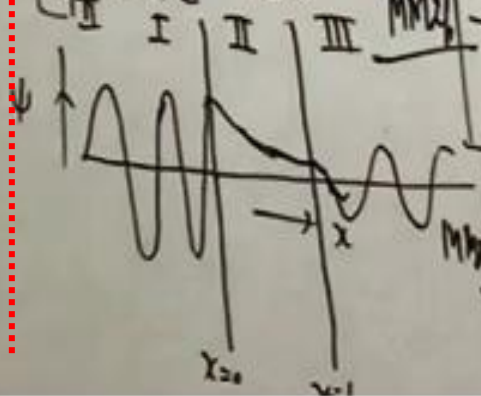
$\kappa_1 A - \kappa_1 B = -\kappa_2 C + \kappa_2 D$
 $C e^{-\kappa_2 L} + D e^{\kappa_2 L} = F e^{\kappa_1 L}$
 $-\kappa_2 C e^{-\kappa_2 L} + \kappa_2 D e^{\kappa_2 L} = \kappa_1 F e^{\kappa_1 L}$

$\psi_I = A e^{\kappa_1 x} + B e^{-\kappa_1 x}$ (I.W. R.W.)
 and $\psi_{III} = F e^{\kappa_1 x} + G e^{-\kappa_1 x}$ Here $G=0$
 $\psi_{II} = C e^{-\kappa_2 x} + D e^{\kappa_2 x}$

Solving for $\frac{A}{F} = \left(\frac{1 + \frac{\kappa_2}{\kappa_1}}{2} \right) e^{(\kappa_1 + \kappa_2)L}$ $\kappa_2 L \gg 1$
 $\frac{A}{F} = \left(\frac{1 - \frac{\kappa_2}{\kappa_1}}{2} \right) e^{(\kappa_1 + \kappa_2)L}$

In Region I $V=0$ $\frac{\partial^2 \psi_I}{\partial x^2} + \frac{2mE}{\hbar^2} \psi_I = 0$ and $\frac{\partial^2 \psi_{III}}{\partial x^2} + \frac{2mE}{\hbar^2} \psi_{III} = 0$

In Region II $\frac{\partial^2 \psi_{II}}{\partial x^2} + \frac{2m}{\hbar^2}(E-V)\psi_{II} = 0$
 or $\frac{\partial^2 \psi_{II}}{\partial x^2} - \frac{2m}{\hbar^2}(V-E)\psi_{II} = 0$



$T(\text{Transmission probability}) = \frac{F F^*}{A A^*} \frac{v_{III}}{v_I} = \left(\frac{A}{F} \right)^{-2}$
 $T = \frac{16}{4 + \left(\frac{\kappa_2}{\kappa_1} \right)^2} e^{-2\kappa_2 L}$
 $\frac{\kappa_2}{\kappa_1} = \frac{\sqrt{2m(V-E)}}{\sqrt{2mE}}$
 $T \approx e^{-2\kappa_2 L}$
 $\kappa_2 = \frac{\sqrt{2m(V-E)}}{\hbar}$